

A line and a circle; the tangent

Three possible relationships, decided by one comparison — and the property that makes tangents useful.

LESSON 53–54

2 × 40 MIN

GEOMETRY 8, TEMA 55

STAGE 9 · 7.1

LESSON CLOCK

6'

12'

12'

6'

4'

Slide a ruler towards a drawn circle	6 min
Three positions	12 min
The tangent property	12 min
Two tangents from a point	6 min
Homework	4 min

LEARNING OBJECTIVES

- Compare the distance from the centre with the radius to classify a line.
- State and use the tangent–radius property.
- Use the fact that two tangents from one point are equal.
- Construct a tangent to a circle at a given point.

TEXTBOOK

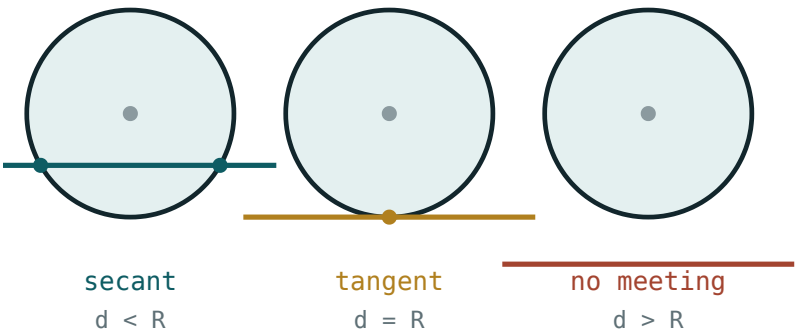
National	Tema 55, pp. 128–131
Cambridge	Learner’s Book pp. 142–148
Workbook	Workbook 7.1

01

Explanation

Three positions of a line

Let d be the distance from the centre O to the line, and r the radius. Everything is decided by comparing them:



A line either misses the circle, touches it once, or cuts it twice — decided by d against r .

COMPARISON	NAME	COMMON POINTS
$d > r$	no name — the line misses	0
$d = r$	tangent	1
$d < r$	secant	2

Remember that “distance from a point to a line” always means the **perpendicular** distance — the shortest one.

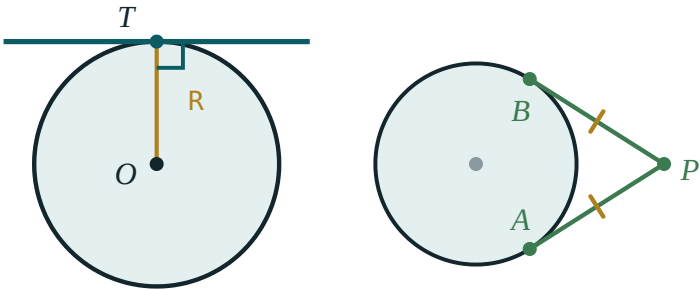
The tangent property

TANGENT AND RADIUS

A tangent is **perpendicular** to the radius drawn to the point of tangency.

$$OT \perp t$$

And conversely: a line through the end of a radius, perpendicular to it, is a tangent.



tangent $\perp$ radius at T two tangents from P are equal

The radius to the point of tangency is perpendicular to the tangent — so a right-angled triangle appears in every tangent problem.

Why: of all the points on a tangent, the point of tangency T is the only one on the circle; every other point is outside, hence further from O . So OT is the shortest distance from O to the line — and the shortest distance is the perpendicular.

The consequence is practical: **every tangent problem contains a right-angled triangle**, so Pythagoras and the trigonometry of Quarter II apply immediately.

Two tangents from one point

EQUAL TANGENTS

From a point P outside a circle, the two tangent segments are **equal**:

$$PA = PB$$

and OP bisects both the angle APB and the angle AOB .

Proof: the triangles OAP and OBP are right-angled at A and B , share the hypotenuse OP , and have equal legs $OA = OB = r$. They are congruent, so $PA = PB$.

To find PA : $PA = \sqrt{OP^2 - r^2}$ — Pythagoras in that right triangle.

02 Worked examples

EXAMPLE 1 A circle has radius 5 cm . A line lies 5 cm from the centre. What is it, and how many points does it share with the circle?

① $d = 5, r = 5$

② $d = r$
The middle case.

③ It is a tangent, meeting the circle at exactly one point.

ANSWER

A tangent; one common point

EXAMPLE 2 From a point P , 13 cm from the centre of a circle of radius 5 cm, a tangent is drawn. Find its length.

① $OT \perp PT$

The tangent property gives a right angle at T .

② $PT^2 = OP^2 - OT^2$

Pythagoras.

③ $= 169 - 25 = 144$

④ $PT = 12 \text{ cm}$

ANSWER

12 cm

EXAMPLE 3 Two tangents from P touch a circle of radius 6 at A and B , and $\angle APB = 60^\circ$. Find OP and PA .

① OP bisects $\angle APB$, so $\angle APO = 30^\circ$

② $\sin 30^\circ = \frac{6}{OP}$

In the right triangle OAP .

③ $OP = \frac{6}{0.5} = 12$

④ $PA = \sqrt{144 - 36} = \sqrt{108} = 6\sqrt{3} \approx 10.4$

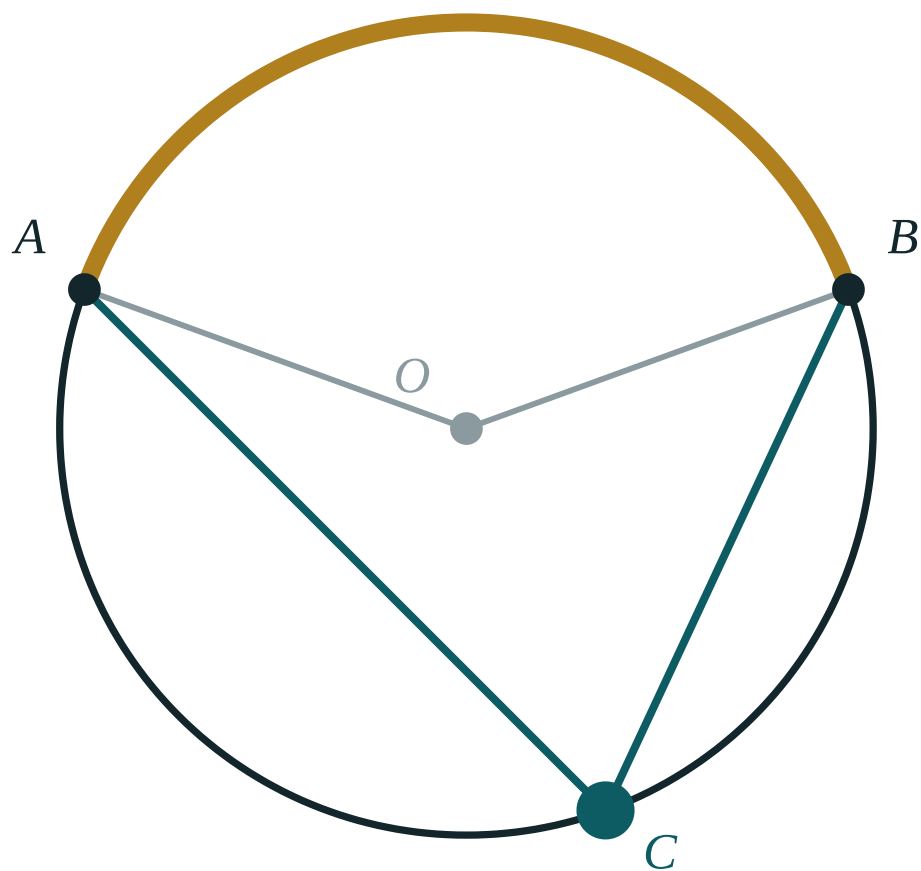
ANSWER

$OP = 12, PA = 6\sqrt{3}$

03 Interactive model

Drag the point round the circle and watch the right angle at the point of tangency.

Radius, chord and tangent



central $\angle AOB$ **140°**

inscribed $\angle ACB$ **70°**

ratio **2**

arc AB **140°**

04 Terminology

Write these on the board at the start. Learners meet the national programme in Uzbek or Russian and the Cambridge papers in English — the three columns are the same idea.

ENGLISH	O'ZBEKCHA	РУССКИЙ
Tangent	Urinma	Касательная
Secant	Kesuvchi	Секущая
Point of tangency	Urinish nuqtasi	Точка касания
Distance to a line	Chiziqqacha masofa	Расстояние до прямой
Perpendicular	Perpendikulyar	Перпендикуляр
Chord	Vatar	Хорда
Radius	Radius	Радиус
External point	Tashqi nuqta	Внешняя точка
Equal tangents	Teng urinmalar	Равные касательные

Stress the words in bold in the explanation above; ask for the Uzbek and Russian back before moving on.

05 Quick check**Check yourself**

If $d < r$, the line is:

a tangent

a secant

outside the circle

a radius

A tangent meets the radius at the point of tangency at:

45°

60°

90°

any angle

A circle of radius 8 and a line 8 from the centre share:

0 points

1 point

2 points

infinitely many

From P with $OP = 10$ and $r = 6$, the tangent length is:

Two tangents from one external point are:

06 Practice · 21 problems

Seven at each level. Set the easy row for everyone, the medium row for the main body of the class, and the hard row for those who finish early.

Easy · warm the idea up

7 PROBLEMS

1. $r = 7, d = 9$. Classify the line.

ANSWER It misses the circle.

2. $r = 7, d = 7$. Classify the line.

ANSWER A tangent.

3. $r = 7, d = 3$. Classify the line.

ANSWER A secant.

4. $OP = 5, r = 3$. Find the tangent length.

ANSWER 4

5. $OP = 13, r = 12$. Find the tangent length.

ANSWER 5

6. What angle does a tangent make with the radius at the point of tangency?

ANSWER 90°

7. PA and PB are tangents with $PA = 9$. Find PB .

ANSWER 9

Medium · the standard question

7 PROBLEMS

1. $OP = 17, r = 8$. Find the tangent length.

ANSWER 15

2. The tangent from P is 24 and $r = 7$. Find OP .

ANSWER 25

3. The tangent from P is 12 and $OP = 15$. Find r .

ANSWER 9

4. A circle of radius 10 has a chord 5 from the centre. Is the chord's line a secant?

ANSWER Yes — $5 < 10$

5. Two tangents from P touch a circle of radius 5 with $\angle APB = 90^\circ$. Find OP .

ANSWER $5\sqrt{2} \approx 7.07$

6. A circle has radius 6 and centre $(0; 0)$. Is the line $y = 6$ a tangent?

ANSWER Yes — its distance from the origin is exactly 6.

7. $OP = 20, r = 16$. Find the tangent length.

ANSWER 12

Hard · stretch and reason

7 PROBLEMS

1. Two tangents from P touch a circle of radius 9 at A and B , with $\angle APB = 60^\circ$. Find PA and the area of $OAPB$.

ANSWER $OP = 18$, $PA = 9\sqrt{3}$, area $81\sqrt{3} \approx 140$

2. A circle of radius 5 is inscribed in a right triangle with legs 12 and 16. Verify that $r = 4$ instead, using area.

ANSWER Hypotenuse 20; $r = \frac{12 + 16 - 20}{2} = 4$, so the radius is 4, not 5.

3. A tangent from P is 15 and $r = 8$. Find OP and $\sin \angle OPA$.

ANSWER $OP = 17$, $\sin \angle OPA = \frac{8}{17}$

4. The line $x = 5$ and the circle $x^2 + y^2 = 25$. How many common points?

ANSWER One — $d = 5 = r$, a tangent at $(5; 0)$.

5. A circle of radius r sits in a corner, touching both arms of a right angle. Its centre is 10 from the corner. Find r .

ANSWER $r\sqrt{2} = 10$, so $r = 5\sqrt{2} \approx 7.07$

6. Two circles of radii 3 and 5 have their centres 10 apart. Find the length of a common external tangent.

ANSWER $\sqrt{100 - 4} = \sqrt{96} \approx 9.8$

7. Explain why the perpendicular from the centre is the shortest distance to a tangent.

ANSWER Every point of a tangent except the point of tangency lies outside the circle, so it is more than r from the centre; the point of tangency is exactly r away, so it is the nearest — and the nearest point on a line is the foot of the perpendicular.

07 Homework

Homework — 5 problems

Geometry 8, Tema 55, pp. 128–131. Mark the right angle in every diagram.

1. $r = 9, d = 4$. Classify the line and say how many common points there are.
2. $OP = 25, r = 7$. Find the tangent length.
3. The tangent from P is 8 and $r = 6$. Find OP .
4. Two tangents from P touch a circle of radius 4 with $\angle APB = 90^\circ$. Find OP and PA .
5. Is the line $y = 7$ a tangent to $x^2 + y^2 = 49$? Justify.